

```
In [1]: import pandas as pd
import seaborn as sd
```

```
In [2]: df= pd.read_csv(r"C:\Users\rehan\Downloads\messy_sales.csv")
```

```
In [3]: df.head(10)
```

```
Out[3]:
```

	order_id	date	product	price	qty	zip
0	1254	04/06/2026	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	05/06/2026	mug	5.50	1	10001
3	1066	05/30/2026	notebook	NaN	1	98101
4	1114	04/06/2026	webcam	45.59	4	90405
5	1280	04/22/2026	pen set	NaN	4	30303
6	1204	04/18/2026	charger	8.41	2	2134
7	1249	2026-05-15	keyboard	75.04	4	90405
8	1037	2026-05-19	mug	9.31	2	60614
9	1177	2026-05-13	mug	10.11	3	90210

```
In [4]: df.shape
```

```
Out[4]: (300, 6)
```

```
In [5]: df.dtypes
```

```
Out[5]: order_id      int64
date              str
product          str
price            float64
qty              int64
zip              int64
dtype: object
```

Three problems

1. The data type of date and zip is wrong
2. Dates aren't properly formatted
3. the zipcode in 6th row is not in the right format

```
In [6]: df.isna().sum() #missing values by column
```

```
Out[6]: order_id    0
        date       0
        product    0
        price     12
        qty        0
        zip        0
        dtype: int64
```

```
In [7]: fill_value = df["price"].mean() #filling missing values
        df["price"] = df["price"].fillna(fill_value)
```

```
In [8]: print(df.duplicated().sum()) #finding and removing duplicates
        df = df.drop_duplicates()
```

8

```
In [9]: df["zip"] = df["zip"].astype(str).str.zfill(5)
```

```
In [10]: df["date"] = pd.to_datetime(df["date"], format="mixed")#Standardize the dates
```

```
In [11]: df[df['qty']<0] #investigating wrong values
```

```
Out[11]:
```

	order_id	date	product	price	qty	zip
202	1140	2026-04-18	webcam	38.69	-5	98101
262	1233	2026-05-09	desk lamp	22.64	-4	10001
297	1025	2026-04-27	keyboard	47.30	-2	02116

Reason for wrong values

I'll keep them as they are and when i'm modeling i'll delete them or try to see if there only a -sign and we can just change the sign.

```
In [12]: df.to_csv("sales_clean.csv", index=False)
```

Cleaning Log

1. Used shape to see the dimentions of the table.
2. Used head(10) to view first 10 rows of the table.
3. Used df.duplicated().sum() to get the number of duplicated records.
4. Removed duplicate records with df.dropduplicates(), it removed 8 rows.
5. Changed zip code format and standarized it.
6. Standarized the date by pd.to_datetime(our column, format='mixed')
7. Looked for wrong values in 'qyt, to clean the data later.

A wrong choice here as have the wrong quatity can show misleading number of sales, if we use can lead the business to spend it resources at the wrong place.

In []: <https://github.com/RehanAhmedSMC/cs82a-data-science-portfolio>